

ARINDAM SINGHA

Lead Engineer — Computer Vision · Robotics Research · Embodied AI · LLM / Agentic AI

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RESEARCH & PRODUCTION PROFILE

Robotics and AI engineer who develops and deploys **deep-learning computer vision algorithms into production**, alongside **13 peer-reviewed publications** in adaptive control, multi-agent coordination, and robotic perception. Experienced across the full path from model research and evaluation to industrial deployment, real-time inference, and production reliability using Docker, Triton, Azure, ROS2, and gRPC. Uniquely positioned at the intersection of **theoretical control design** (Lyapunov stability, RBFNN, graph-theoretic consensus) and **computer vision** (object detection, semantic segmentation, keypoint detection, and multi-camera systems). Seeking a Lead Engineer or Research Scientist role to advance applied research in embodied AI, multi-robot systems, and autonomous perception.

CORE EXPERTISE

Computer Vision:	YOLOv3/v4/v8 object detection, semantic segmentation, keypoint detection, multi-camera vision pipelines, model evaluation, and real-time inference
Control & Robotics Theory:	Adaptive RBFNN control, Lyapunov-based stability analysis, leader-follower consensus, graph-theoretic network topologies, nonholonomic constraints, multi-UAV/UGV coordination
Deep Learning Research:	Transformers, encoder-decoder architectures, architecture comparison, accuracy-latency analysis, digital twin simulation (Webots)
Embodied AI & RL:	MDP formulation, Actor-Critic, PPO, MADDPG/MATD3, ROS2/Gazebo/RViz, MoveIt2, NAV2, URDF, TF2, DDS
MLOps & Reproducibility:	Docker, Triton Inference Server, Azure Cloud, CI/CD, MLflow, DVC, Databricks, gRPC microservices
Agentic AI:	LangChain, LangGraph, RAG, Agentic RAG, MCP protocols for automated test-case generation
Programming:	Python, MATLAB, C/C++
Languages:	English (Fluent, professional), Hindi (Fluent), Bengali (Native)

RESEARCH EXPERIENCE

Lead Engineer — Computer Vision & AI

Jul 2024 – Present

Magna Steyr India, Pune, India

- Develop and deploy deep-learning computer vision algorithms into **24/7 production systems**, including YOLOv8 multi-class defect detection and semantic segmentation with >87% mAP.
- Own the path from model experimentation and validation to containerized release and **real-time production inference**, using Docker and Triton Inference Server for latency-constrained vision pipelines.
- Lead migration of a microservice vision platform to **ROS2-based real-time architecture**, defining middleware interfaces (DDS, gRPC) for deterministic sensor-to-actuator pipelines.
- Develop an **Agentic-RAG system** (LangGraph + MCP) for automated requirements-to-test-case generation, demonstrating LLM-based reasoning in safety-critical engineering workflows.
- Led keypoint detection benchmarking using RSN, MSPN, and YOLO-NAS for pose estimation and component alignment tasks.

Senior Engineer — Computer Vision & AI

Feb 2022 – Jun 2024

Magna Steyr India, Pune, India

- Built a **Webots digital twin** of the Mobileye Q3 camera with a ROS2 bridge, enabling pre-deployment algorithm validation and reducing field-calibration time by 25%.
- Designed distributed gRPC microservices for multi-camera vision processing, analyzing communication latency and fault-tolerance trade-offs.

Senior Research Fellow

Jan 2018 – Dec 2021

National Institute of Technology Sikkim, India

- Designed an **adaptive RBFNN-based control framework** for multi-UAV formation control under external disturbances, with rigorous Lyapunov stability proofs and bounded-error guarantees.
- Developed leader-follower formation controllers for quadrotor UAVs under **directed and undirected graph topologies** (fixed and time-varying adjacency), analyzing consensus convergence via algebraic graph theory.
- Validated controllers in Gazebo/ROS2 simulations tests under wind gust disturbances.
- Published **8 peer-reviewed journal articles** (Springer, Elsevier) and presented at international conferences on networked robotic systems.

Junior Research Fellow

Jan 2016 – Dec 2017

National Institute of Technology Sikkim, India

- Developed neural-network path-planning algorithms (A^* + NN heuristic) for grid-based UGV navigation in partially known environments.
- Built an Extended Kalman Filter framework for multi-UAV trajectory tracking with IMU/GPS sensor fusion and covariance analysis.

SELECTED PUBLICATIONS

A. Singha et al., "Adaptive Neural Network based Quadrotor UAV Formation Control under External Disturbances," *Aerospace Science and Technology* (Springer), 2024.

A. Singha et al., "Lyapunov based Trajectory Tracking Controller for a Quadrotor UAV with Nonholonomic Constraints," *e-Prime – Advances in Electrical Engineering, Electronics and Energy* (Elsevier), 2024.

A. Singha et al., "UAV Formation Control under Fixed and Variable Adjacency Based Directed Network Topologies," *Journal of Scientific and Industrial Research* (CSIR-NIScPR), 2022.

A. Singha et al., "Leader-follower Based Formation Controller Design for Quadrotor UAVs," *Transactions of the Indian National Academy of Engineering* (Springer), 2022.

Full list of 13 peer-reviewed publications available on ResearchGate.

EDUCATION

Ph.D., Intelligent Networked Robotic Systems 2016 – 2021

National Institute of Technology Sikkim, India

Dissertation: Intelligent Networked Robotic Systems

M.E., Electronics, Instrumentation & Control 2012 – 2014

Thapar University, India

B.Tech, Applied Electronics and Instrumentation Engineering 2008 – 2012

Abacus Institute of Engineering and Management (MAKAUT), India

RESEARCH PROJECTS

Agentic Requirements Engineering (LLM + RAG + MCP) May 2025 – Present

Multimodal document ingestion pipeline with vector embedding, retrieval-augmented generation, and LangGraph-based agent orchestration for automated test-case synthesis from unstructured requirements.

Motion Planning Benchmarking — 6-DoF Manipulator Jul 2024 – Jan 2025

Benchmarked RRT/RRT-Connect planners in ROS2/MoveIt2; achieved 40% path-optimization improvement via heuristic biasing and collision-geometry caching.

AWARDS & GRANTS

- Visvesvaraya PhD Scholarship — Digital India Corporation, Ministry of Electronics & IT, Govt. of India
- MHRD GATE Scholarship — Ministry of Human Resource Development, Govt. of India
- Team of the Month — Magna Steyr India (Oct 2023, Apr 2025)
- Employee of the Month — Magna Steyr India (Dec 2022)

Mobility & Work Authorization

Open to relocation across EU countries. Require employer-sponsored EU Blue Card/work visa. Available for immediate relocation.